

Remembering Primates

The 8 Taxonomic Ranks

Domain, Kingdom, Phylum, Class, Order, Family, Genus, Species

Dear King Philip Came Over From Good Spain

But relevant here are five trick ranks!

Suborder (Smaller sections of Order)

Infraorder

Parvorder

Superfamily (Between order and family)

Sub-family

Tribe (between Family and Genus)

Strepsirrhini and Haplorrhini

Strepsirrhini and Haplorrhini are the first large divide. If you remember that humans are in Haplorrhini, you can remember what traits Haplorrhini have versus Strepsirrhini based on the traits you have! Most Strepsirrhini traits are also characteristic of primate ancestral traits: remember the Strepsirrhini traits first and then remember at what point Haplorrhini primates diverge from them.

Strepsirrhini (wet nosed) versus Haplorrhini (dry-nosed)

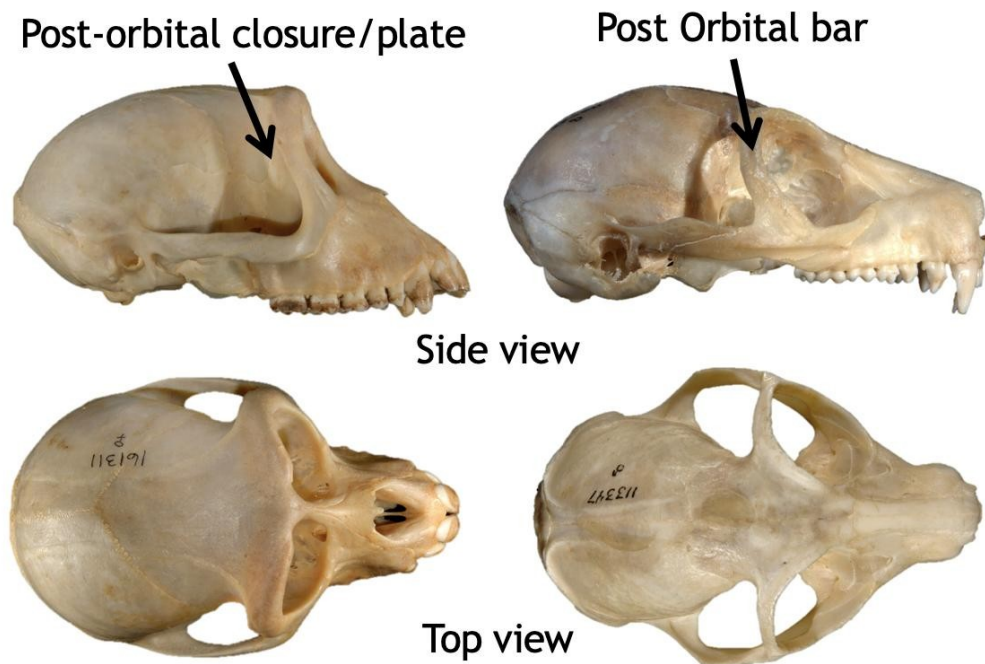
S= Shiny (wet), H = human (humans are a part of haplorrhini)

Strepsirrhini are characterized by:

- Post-orbital bar: the eye sockets are not fully closed behind the eyes, and have just a ring/bar going around the back. Human skulls in contrast are fully closed; imagine a human skull and recall that you cannot see past the eyes because the bones are fully blocking the view. Strepsirrhini are the opposite of that.
- Unfused frontal bone: Two separate bones (hence unfused) forming the forehead. You only have one!
- Unfused mandibular symphysis: Two lower halves of the lower jaw not used. You don't have this.
- Nails and Toilet Claws: Specialized grooming claw on second toe. You don't have this.
- Dental comb: Lower front teeth used for grooming. You don't have this.
- Nocturnal: Strepsirrhini tend to mainly go out at night. Most of you don't.
- Dental formula 2-1-3-3: Imagine splitting your skull in half and counting the teeth in your mouth. You would most likely count two front teeth, one canine, two premolars, and three back molars. But for most strepsirrhini, they have 2-1-3-3, or one more premolar.

The lower ranks in Strepsirrhini are then easy to remember. For our purposes there is only one infraorder, Lemuriformes, and two super families, Lemuroidea and Lorisodea. Lorises are exemplar

Strepsirhini and are in Africa and Asia. However, Lemurids differ a little bit: some of them are diurnal (out during the day) rather than nocturnal, and some have a 2-1-2-3 dental formula.



Tarsiiformes

Most Haplorhini don't have these features, EXCEPT for the weird infraorder: Tarsiiformes. That is why they are so controversial and don't fit easily between Strepsirhini and Haplorhini.

Tarsiiformes still have:

- Unfused mandibular symphysis, toilet claw, nocturnal lifestyle.
- BUT! They do have partial post-orbital septum. Note it is only partial, not full.
- They are the ONLY Haplorhini to have unfused mandibles and toilet claws.

Think of them as in between Strepsirhini and Haplorhini, even though they are under Haplorhini. They are also only left in Southeast Asia.

Mnemonic: Tarsiers are weird and drift in the SEA.

- If you remember that Tarsiiformes are weird you will recall why, and that they are in SEA (Southeast Asia).

Anthropoidea

One more step closer towards humans, and when the anthro- starts appearing everywhere. All four of the key derived features that define higher primates appear simultaneously at this level. Derived traits introduced:

- Fused frontal bone (two bones merge into one solid forehead bone)

- Fused mandibular symphysis (lower jaw halves permanently fuse into one rigid bone)
- Full post-orbital closure (the eye orbit is completely enclosed in bone)
- Post-orbital septum (a bony wall fully separates the eye socket from the jaw muscles behind it)
- Disappearance of dental comb and toilet claw

In short, all the things you have, except for dental formula, narrow nasal septum, molar pattern, etc...

Platyrrhini and Catarrhini

The only parvorder to remember. The main divide is over whether your nostrils go flat or down, as well as dental formula: 2-1-2-3 starts appearing in both, but Catarrhini (the human side) is now ALL 2-1-2-3.

Mnemonic:

Platy- is like platypus' flat, broad bill. Cata- means to go down, like catastrophe (literally meaning down-turn in Greek). Your nose goes down, so you're on the catarrhini side!

Cebonoidea

Superfamily on the Platyrrhini (not you) side. These are all new world monkeys, in contrast to cercopithecoidea, which are old world monkeys. Apes like you are old world too.

Cercopithecoidea and Hominoidea

This one separates the Old World Monkeys from the Apes, greater and lesser. The crucial divide is over molar patterns: old world monkeys have bilophodont molar patterns (four cusps in two ridges), apes, like you, have Y-5 molar pattern instead. They also differ in the following features:

- Ischial collosities: Hardened sitting pads on the buttocks. Some of you may not have buttocks don't have hair, but not in a way that makes them special and separate from the rest of your legs, so you don't have ischial collosities (humans actually make up for the lack of them when sitting by having more fat on their behinds). Thus you can remember it's an old world monkey trait EXCEPT for gibbons and siamangs (lesser apes), which retain them.
- Loss of tail: You don't have an external tail (although you do have a leftover tailbone). Thus remember retaining the tail as a cercopithecoidea trait.
- Y-5 Molar Pattern: Apes start having the Y-5 molar pattern, which is suited to an omnivorous diet. Note that humans and other primates actually have a lot of dental pattern variation (Y-5, extra 6th cusp) etc, but Y-5 is the standard. Don't get tricked by a question on the exam that might state that ALL humans have Y-5 uniformly though.
- Brachiator anatomy: These are modifications in the shoulder joints and blades and chest that allows (some) apes to swing with long arms from branch to branch (remember: **BRACH**iation = kinda looks like BRANCH). The lesser apes are brachiators, although the great apes shift away from them.

Hylobatidae and Hominidae

The second to last division before humans, between lesser and great apes. There are only two species under the 1 genus of lesser apes: Gibbons and the lesser known Siamang. Only they have ischial collosities under Hominoidea. There will probably be a trick question saying that gibbons are the only ape to have IC; remember that siamangs exist too!

Remember that siamangs and gibbons only exist in Southeast Asia, just like Tarsiers. They're also kinda weird too because they're apes but lesser.

For the great apes, remember their locomotor strategies:

GiPoGoCH: Gibbons (and Siamangs) > Orangatang > Gorillas > Chimps > Humans for the order of brachiation to fist walking to knuckle walking to bipedalism. Location wise remember this also correlates to island apes to African apes to us, which come out from Africa. Orangatang is the only great ape NOT in Africa.

All Latin meanings:

Primate: *primus* (first)

Strepsirrhini: *strepsi-rhini* (twisted nose)

Haplorhini: *haplo-rhini* (simple nose)

Lemuriformes: *lemur-formes* (ghost shaped)

Lorisoidea: *lories* (clown)

Tarsiiformes: *tarsos-formes* (ankle-shaped)

Anthropoidea: *anthropo-oidea* (human-like)

Platyrrhini: *platy-rhini* (flat nose)

Catarrhini: *kata-rhini* (downward nose)

Cercopithecoidea: *kerkos-pithecus* (tail ape)

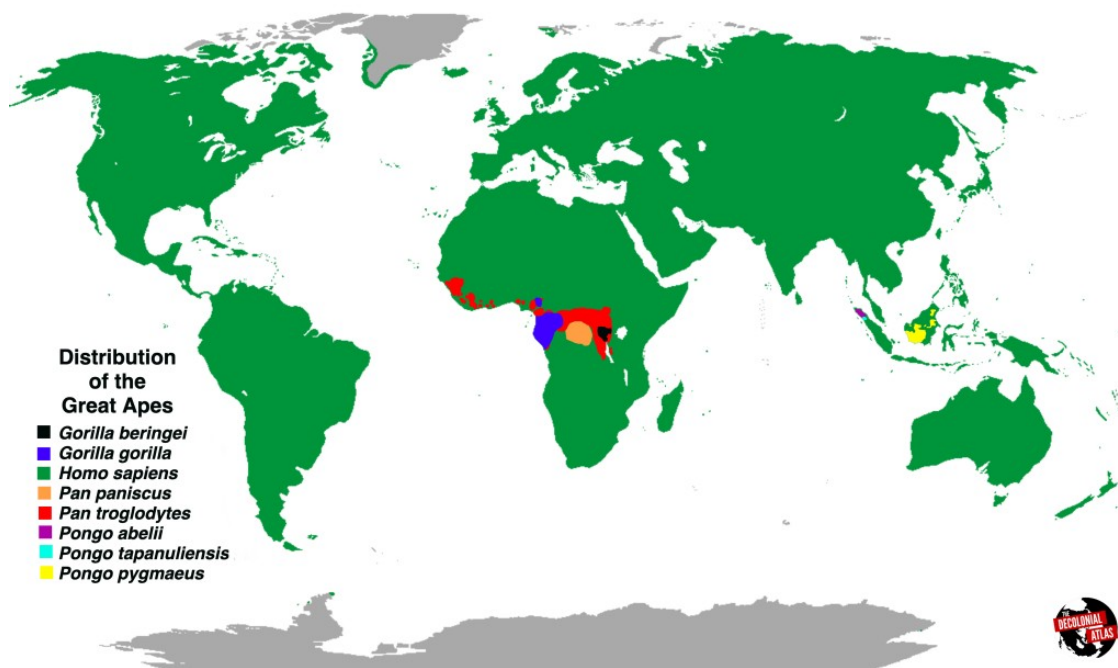
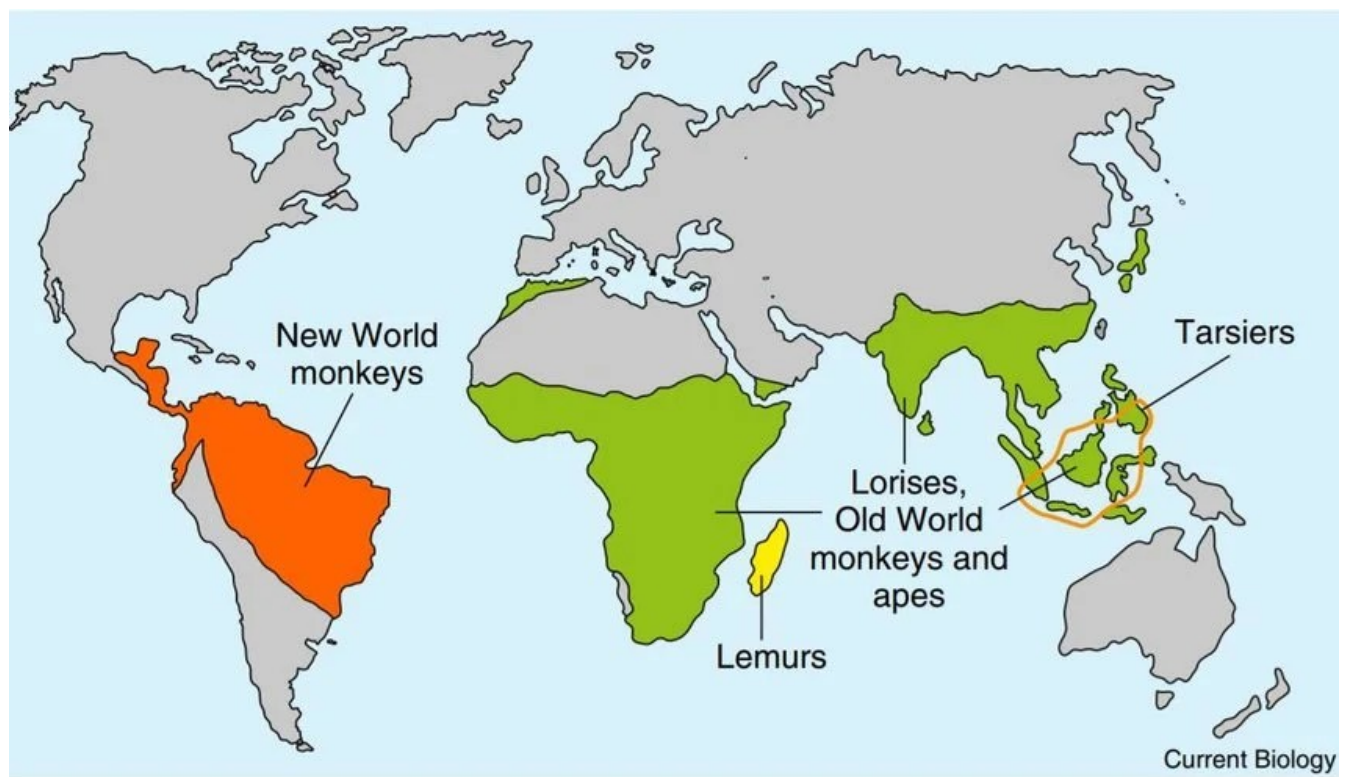
Hominoidea: *homo* (man)

Hylobatidae: *hylo-bates* (tree walker)

Ponginae: *pongo* (Kongo for sleep)

Gorillinae: *gorillai* (tribe of hairy people)

Panini: *Pan* (Greek god of wild nature)



PRIMATES

